Your grade: 100%

Your latest: **100%** • Your highest: **100%** • To pass you need at least 80%. We keep your highest score.

1. What is a key advantage of using parallel I/O over traditional serial I/O in MPI applications? 1 / 1 point

- It's easier to implement
- It allows many processes to communicate directly with the storage system
- It reduces the total amount of data written
- It automatically optimizes file system performance

Nice work

Correct, this alleviates bottlenecks and improves performance.

2. What is a potential drawback of using small I/O operations in parallel file systems? 1 / 1 point

- Increased network latency
- Higher energy consumption
- Wasted space due to file system block size
- Reduced data integrity

Nice work

Correct, small files still occupy a full block, leading to inefficient space usage.

3. What is "striping" in the context of parallel file systems? 1 / 1 point

- A method of data compression
- A way of distributing file data across multiple disks
- A technique for error checking
- A process of file encryption

Nice work

Correct, striping spreads file data across available disks to improve I/O performance.

4. In MPI I/O, which operations are typically collective? 1 / 1 point

- MPI_File_Open and MPI_File_Close
- MPI_File_Seek and MPI_File_Write
- Only MPI_File_Read
- All MPI I/O operations are collective

Nice work

Correct, these operations require all ranks in the communicator to participate.

5. What is the primary purpose of HDF5 in the context of parallel computing? 1 / 1 point

- To replace MPI entirely
- To provide an easier interface for parallel I/O than raw MPI I/O
- To optimize network communication
- To manage process synchronization

Nice work

Correct, HDF5 simplifies parallel I/O operations compared to lower-level MPI I/O.

6. In HDF5, what is the relationship between a file and its root group? 1 / 1 point

- They are separate entities
- The file object represents the root group
- The root group contains multiple files
- There is no relationship between files and groups in HDF5

Nice work

Correct, the file object in HDF5 represents the root of the hierarchical structure.

7. Which mode should be used when opening an HDF5 file for both reading and writing without overwriting existing data? 1 / 1 point

- 'w'
- 'r'
- 'r+'
- 'a'

Nice work

Correct, this mode allows both reading and writing without overwriting the existing file.

8. Why is it important to consider the memory layout when performing parallel I/O? 1 / 1 point

- To optimize CPU cache usage
- To ensure data consistency across processes
- To minimize inter-process communication
- To align data access with file system structure

Nice work

Correct, proper alignment can significantly improve I/O performance in parallel systems.

9. In the context of HDF5, what is the primary difference between datasets and groups? 1 / 1 point

- Datasets can only store numerical data, while groups can store any type of data
- Datasets are array-like containers of data, while groups are like directories
- Groups can only contain datasets, not other groups
- Datasets are used for parallel I/O, while groups are for serial I/O only

Nice work

Correct, this analogy is used in the lecture to explain the difference.

10. When using h5py for parallel I/O, which driver should be specified when creating a file? 1 / 1 point

- MPI
- MPIO
- HDF5
- Parallel

Nice work

Correct, the lecture mentions specifying the MPIO driver for parallel access.